Part A

Motivation

In order to match hospital staffing and capacity with patient demand, STADIOcare Group needs improved forecasting. Predictive analysis is being used to enhance planning and resource utilization due to declining occupancy, longer patient stays, and increased labor costs.

Problem statement

Forecasting patient demand, bed utilization, and length of stay across its 44 hospitals is a challenge for STADIOcare. Over the course of two years, the average duration of stay climbed from 3.9 to 4.2 days, occupancy decreased from 71.5% to 69.2%, and agency and overtime expenditures jumped from R610 million to R890 million. Managers, employees, and maybe patients are impacted by these difficulties. Data on admissions, appointments, billing, and human resources can be examined; however, evaluation of the data's quality and historical coverage is necessary. The goal of the project is to create predictive models that facilitate proactive workforce and capacity planning.

Model choices

Using a seasonal naïve model as a baseline, SARIMAX will be taken into consideration for forecasting admissions and occupancy. The length of stay will be predicted using Random Forest regression, with linear regression serving as a comparison. Root mean squared error, mean absolute error, and temporal data splits will be used to assess the models.

Statistical analysis

In addition to summarizing admissions, occupancy, duration of stay, and labor costs, analysis will evaluate the quality of the data. We'll look into trends, seasonal patterns, and hospital-specific variations. Relationships between operational metrics will be examined using regression and correlation. The provided figures indicate descriptive changes; once comprehensive data are available, statistical significance and model performance will be evaluated.